

Splunk Enterprise Alert Triage Brute Force Attack Example

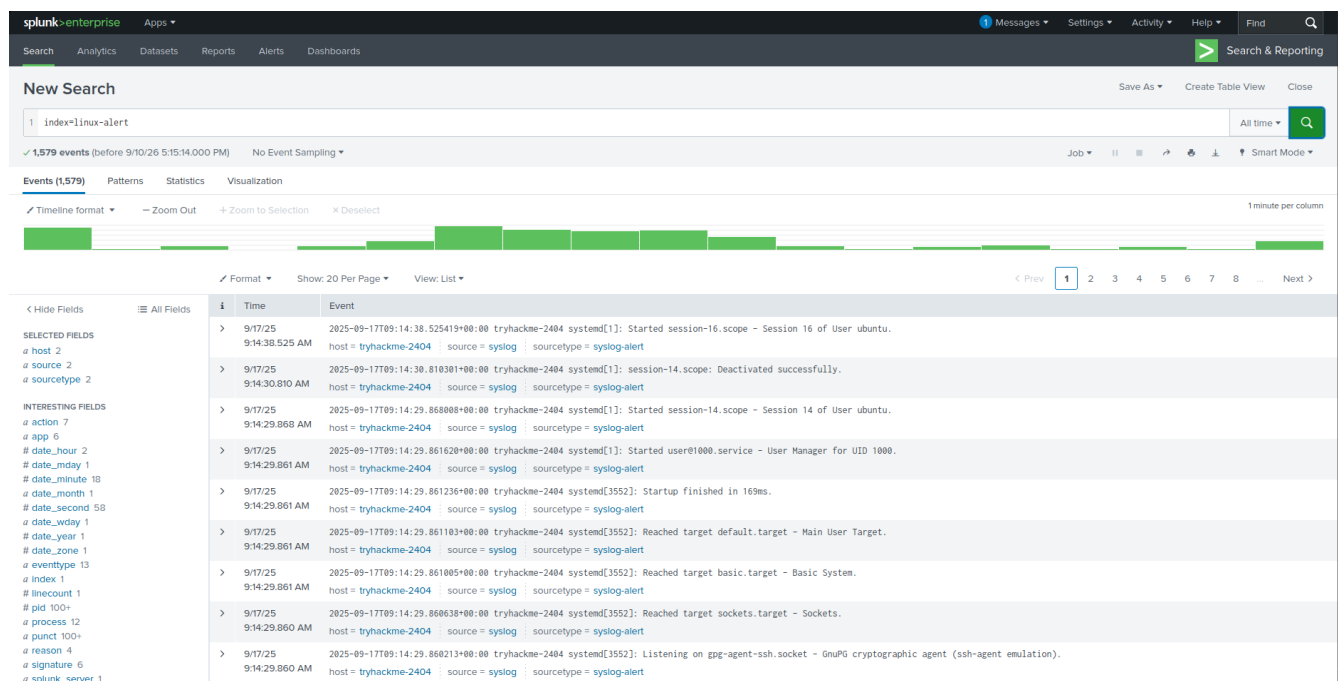
Scenario

In this example, I have just been given the job as a SOC Analyst as a Managed Security Service Provider (MSSP). The following was detected as a brute force attack with the alert details as follows:

- **Alert Name:** Brute Force Activity Detection
- **Time:** 17/09/2025 9:00:21 AM
- **Source IP:** 10.10.242.248
- **Target Host:** tryhackme-2404

My job was to investigate this activity and decide whether it should be considered suspicious.

As a SOC analyst, my first task was to open the logs for this particular activity by opening the following query in Splunk Enterprise **index=linux-alert**:



I then began the investigation by focusing on the alert data. The two fields that were of interest was the **Target Host Time & Source IP**. By reviewing the IP address, I can see that the attacker was already inside of the organization's network, possibly compromising the VPN or gained access through another way.

By using the following command in Splunk, I narrowed down the events (1,579 in total) down to 543 events that were compromised by the attacker:

```
index="linux-alert" sourcetype="linux_secure" 10.10.242.248
| search "Accepted password for" OR "Failed password for" OR "Invalid user"
| sort + _time
```

New Search

```

1 index="linux-alert" sourcetype="linux_secure" 10.10.242.248
2 | search "Accepted password for" OR "Failed password for" OR "Invalid user"
3 | sort -_time

```

✓ 543 events (before 9/10/26 5:15:27:000 PM) No Event Sampling

Events (543) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 minute per column

Format Show: 20 Per Page View: List

< Hide Fields	All Fields	i	Time	Event
		>	9/17/25 9:00:18.980 AM	2025-09-17T09:00:18.980749+00:00 tryhackme-2404 sshd[2798]: Invalid user emma.johnson from 10.10.242.248 port 34808 host = ce-splunk source = auth.log sourcetype = linux_secure
		>	9/17/25 9:00:18.985 AM	2025-09-17T09:00:18.985437+00:00 tryhackme-2404 sshd[2798]: Disconnected from invalid user emma.johnson 10.10.242.248 port 34808 [preauth] host = ce-splunk source = auth.log sourcetype = linux_secure
		>	9/17/25 9:00:19.259 AM	2025-09-17T09:00:19.259010+00:00 tryhackme-2404 sshd[2800]: Invalid user emma.johnson from 10.10.242.248 port 38346 host = ce-splunk source = auth.log sourcetype = linux_secure
		>	9/17/25 9:00:21.045 AM	2025-09-17T09:00:21.045113+00:00 tryhackme-2404 sshd[2800]: Failed password for invalid user emma.johnson from 10.10.242.248 port 38346 ssh2 host = ce-splunk source = auth.log sourcetype = linux_secure
		>	9/17/25 9:00:22.733 AM	2025-09-17T09:00:22.733074+00:00 tryhackme-2404 sshd[2800]: Connection closed by invalid user emma.johnson 10.10.242.248 port 38346 [preauth] host = ce-splunk source = auth.log sourcetype = linux_secure
		>	9/17/25 9:00:29.361 AM	2025-09-17T09:00:29.361767+00:00 tryhackme-2404 sshd[2802]: Invalid user emma.johnson from 10.10.242.248 port 53424 host = ce-splunk source = auth.log sourcetype = linux_secure
		>	9/17/25 9:00:29.367 AM	2025-09-17T09:00:29.367550+00:00 tryhackme-2404 sshd[2802]: Disconnected from invalid user emma.johnson 10.10.242.248 port 53424 [preauth] host = ce-splunk source = auth.log sourcetype = linux_secure
		>	9/17/25 9:00:29.632 AM	2025-09-17T09:00:29.632568+00:00 tryhackme-2404 sshd[2804]: Invalid user emma.johnson from 10.10.242.248 port 53438 host = ce-splunk source = auth.log sourcetype = linux_secure
		>	9/17/25 9:00:29.632 AM	2025-09-17T09:00:29.632568+00:00 tryhackme-2404 sshd[2804]: Invalid user emma.johnson from 10.10.242.248 port 53438 host = ce-splunk source = auth.log sourcetype = linux_secure

By using this instance I have not found any signs of brute force attempts as of yet. So to continue further I had to narrow down the users that had an unusual amount of login attempts made. So with the following instance written, I have found the user John Smith (john.smith) that had around 500 login attempts under the statistics tab.

```

index="linux-alert" sourcetype="linux_secure" 10.10.242.248 "Failed Password"
| stats count by src_ip user_name
| sort -count

```

New Search

```

1 index="linux-alert" sourcetype="linux_secure" 10.10.242.248 "Failed Password"
2 | stats count by src_ip user_name
3 | sort -count

```

✓ 508 events (before 9/10/26 6:40:52:000 PM) No Event Sampling

Events Patterns **Statistics (4)** Visualization

Show: 20 Per Page Format Preview: On

src_ip	user_name	count
10.10.242.248	john.smith	500
10.10.242.248	david.miller	3
10.10.242.248	sarah.williams	3
10.10.242.248	emma.johnson	2

I now know the username John Smith credentials were compromised, so I entered his username into the field. I have come to find out that the brute force attack happened in the duration of 5 minutes

```

index="linux-alert" sourcetype="linux_secure" user_name=john.smith 10.10.242.248 "Failed Password"
| bin span=1m _time | stats count values(user_name) as username by src_ip _time
| sort -_time

```

splunk>enterprise Apps Messages Settings Activity Help Find Search & Reporting

Search Analytics Datasets Reports Alerts Dashboards

New Search Save As Create Table View Close

index="linux-alert" sourcetype="linux_secure" user_name=john.smith 10.10.242.248 "failed Password"

2 | bin span=1m _time | stats count values(user_name) as username by src_ip _time

3 | sort -_time

✓ 500 events (before 9/10/26 7:21:06.000 PM) No Event Sampling

Jobs Visualization Smart Mode

Events Patterns **Statistics (6)** Visualization

Show: 20 Per Page Format Preview: On

src_ip	_time	count	username
10.10.242.248	2025-09-17 09:06:00	25	john.smith
10.10.242.248	2025-09-17 09:05:00	107	john.smith
10.10.242.248	2025-09-17 09:04:00	98	john.smith
10.10.242.248	2025-09-17 09:03:00	90	john.smith
10.10.242.248	2025-09-17 09:02:00	135	john.smith
10.10.242.248	2025-09-17 09:01:00	45	john.smith

Knowing what I know now, I have used John Smith to look up the failed attempts and managed to look up “the actual” person who used his credentials, which the user is “root” using the following instance in the events information:

john.smith action!=failure

splunk>enterprise Apps Messages Settings Activity Help Find Search & Reporting

Search Analytics Datasets Reports Alerts Dashboards

New Search Save As Create Table View Close

john.smith action!=failure

✓ 14 events (before 9/10/26 7:53:30.000 PM) No Event Sampling

Jobs Visualization Smart Mode

Events (14) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 minute per column

Time	Event
9/17/25 9:12:33.101 AM	2025-09-17T09:12:33.101140+00:00 tryhackme-2484 sshd[3327]: pam_unix(sshd:session): session closed for user john.smith
9/17/25 9:12:33.099 AM	2025-09-17T09:12:33.099689+00:00 tryhackme-2484 sshd[3385]: Disconnected from user john.smith 10.10.242.248 port 53244
9/17/25 9:12:10.763 AM	2025-09-17T09:12:10.763798+00:00 tryhackme-2484 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by john.smith(uid=0)
9/17/25 9:11:28.984 AM	2025-09-17T09:11:28.984986+00:00 tryhackme-2484 su[33993]: pam_unix(su:session): session opened for user root(uid=0) by john.smith(uid=0)

Summary

This is an example of how Splunk Enterprise can be used to investigate, examine and expose vulnerabilities when it comes to brute force attacks. Writing the right instances and executing the code results in the attacker’s information being exposed and would remediate a plan with the risk management team and have better MFA practices with personnel.